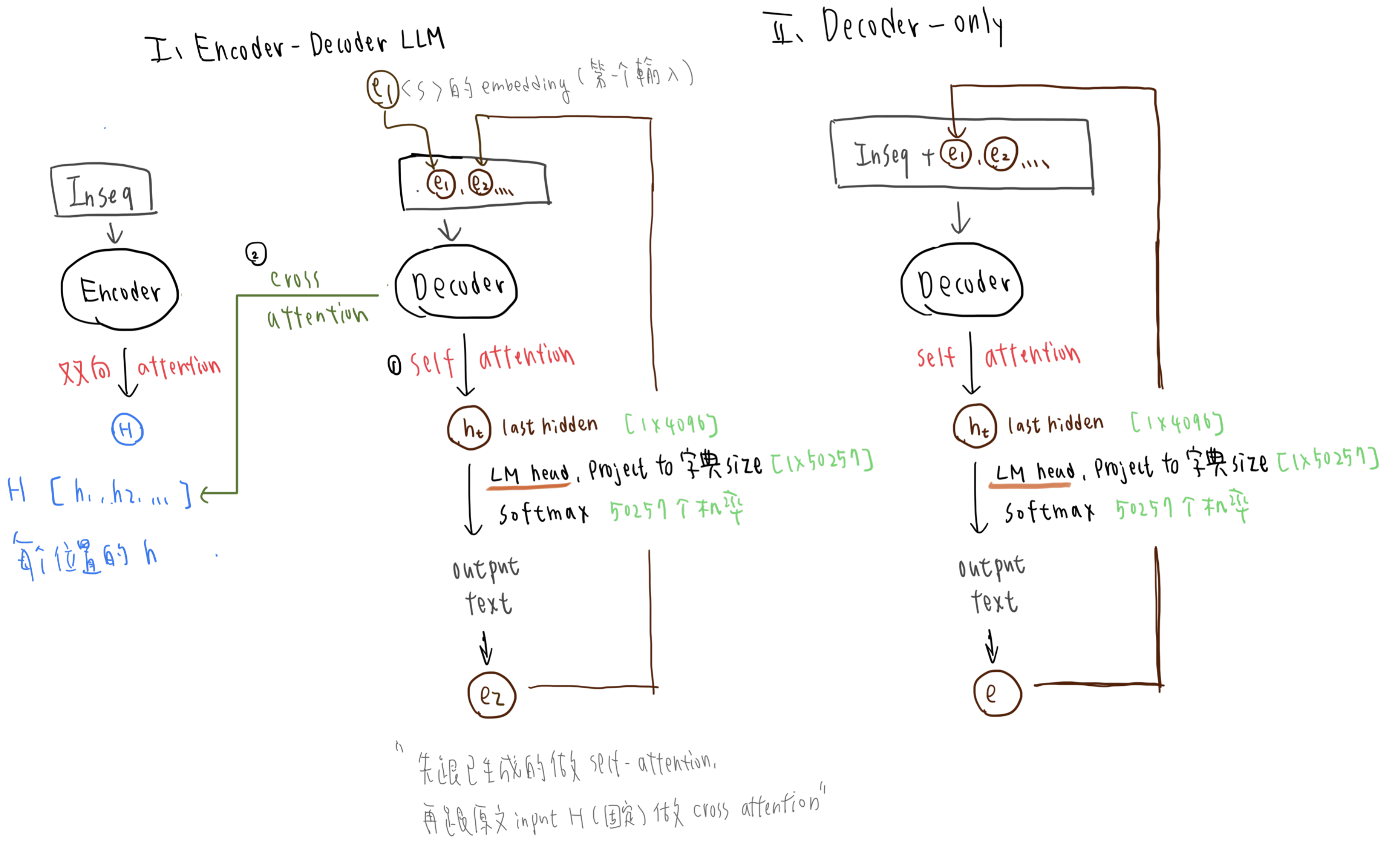
Input text seq.

↓ tokenization + embedding look up.

$\boxed{\text{Inseq}}$ embedding seq



I、Encoder - Decoder LLM
e1 <s> 的 embedding（第一个輸入）
Inseq
Encoder
双向 attention
H
H [h1, h2, ...]
每个位置的 h
② cross attention
e1, e2, ...
Decoder
① self attention
ht last hidden [1x4096]
LM head, Project to 字典 size [1x50257]
softmax 50257个机率
output text
e2
II、Decoder - only
Inseq + e1, e2, ...
Decoder
self attention
ht last hidden [1x4096]
LM head, Project to 字典 size [1x50257]
softmax 50257个机率
output text
e

"先跟已生成的做 self-attention,
再跟原文 input H（固定）做 cross attention"

## • Decoder-only

↳ Train: <s> $\boxed{T_1}$ $\boxed{T_2}$ $\boxed{T_3}$ </s>

（假設只有1層）

$e_1$ $e_2$ $e_3$ $e_4$ $e_5$

$\overset{Q}{e_1} \rightarrow \boxed{\overset{K}{e_1}} \rightarrow \overset{V}{h_1}$ (−) $e_2$ Loss

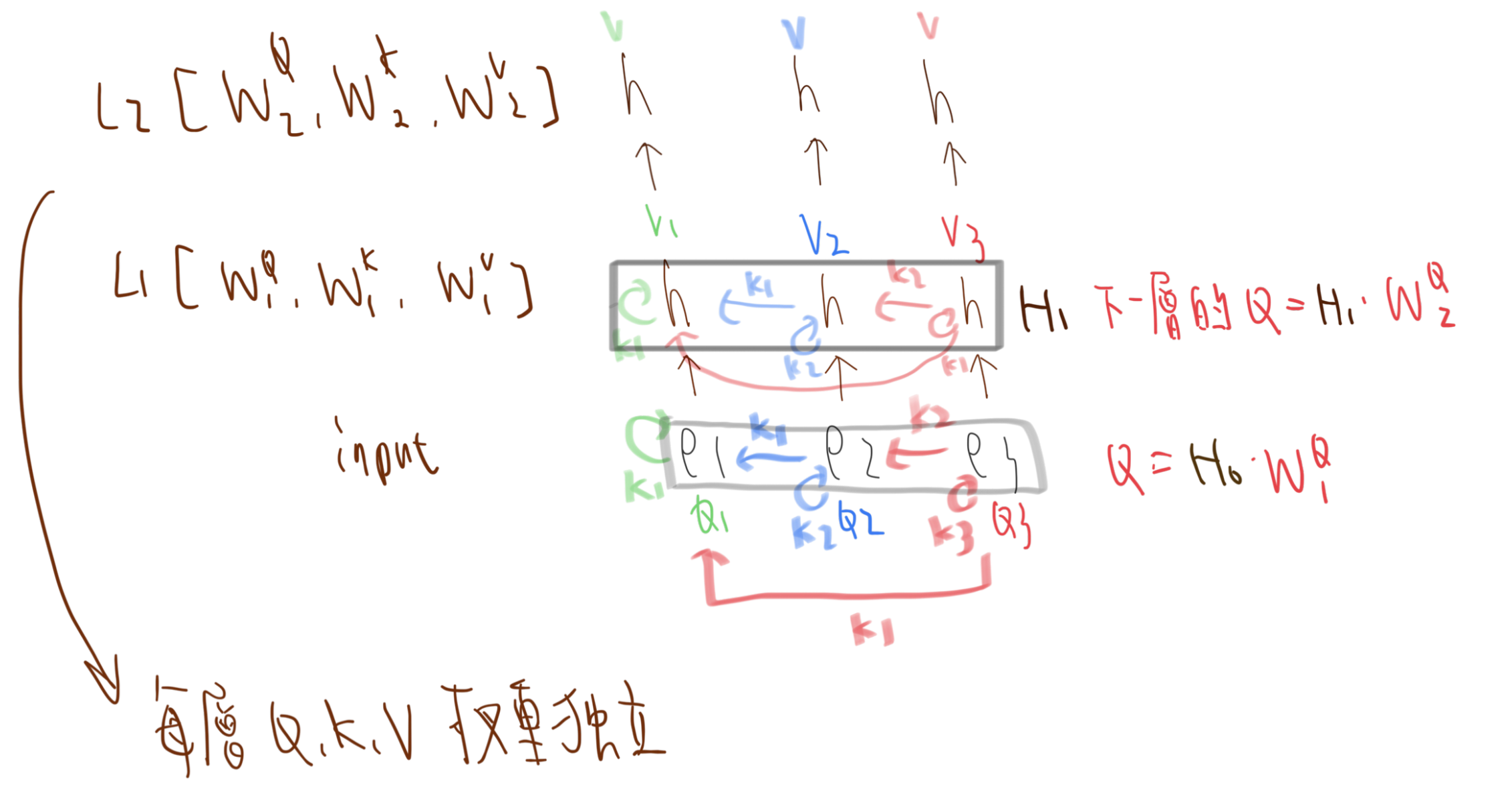
$\overset{Q}{e_2} \rightarrow \boxed{\overset{K}{e_1}, \overset{K}{e_2}} \rightarrow \overset{V}{h_2}$ (−) $e_3$ Loss

$\overset{Q}{e_3} \rightarrow \boxed{\overset{K}{e_1}, \overset{K}{e_2}, \overset{K}{e_3}} \rightarrow \overset{V}{h_3}$ (−) $e_4$ Loss

⋮

↳ Inference: $\underbrace{<s>\ \boxed{T_1}\ \boxed{T_2}}_{\text{input}}$ $\underbrace{\boxed{T_3}}_{\text{output}}$

$e_1$ $e_2$ $e_3$ $e_4$



L2 [W₂^Q, W₂^K, W₂^V]
L1 [W₁^Q, W₁^K, W₁^V]
input
h h h
V1 V2 V3
K1 K2 K1
e1 e2 e3
Q1 K2 Q2 K3 Q3
K1
H1 下一層的 Q = H1 · W₂^Q
Q = H0 · W₁^Q
每層 Q, K, V 权重独立

## • Attention:

↳ Q、K、V：Q、K、V 各自有个已訓練好的权重矩陣（$W^Q, W^K, W^V$）

K：token 的重要特徵

V：token 的完整内容

$h_1, h_2, h_3, \ldots, h_t$（III）

seq: $e_1, e_2, e_3, \ldots, e_t \rightarrow h_{t+1}$ 的產生：

$K\ [k_1\ k_2\ k_3 \ldots k_{t-1}]$

$V\ [v_1\ v_2\ v_3 \ldots v_{t-1}]$

1x4096

I、$Q_{new} : X \cdot W^Q$

$K_{new} : X \cdot W^K$

$V_{new} : X \cdot W^V$

[Store in Cache]

下个 output 用

II、Attention

$\text{softmax}(Q_{new} \times K)$

$A : [\overset{A_1}{0.1}, \overset{A_2}{0.8}, \overset{A_3}{0.01}, \ldots]$

前面 seq 的 token，各自重要度

III、Weighted sum

$A_1 \times V_1 + A_2 \times V_2 \ldots A_{t-1} \times V_{t-1} = h_{t+1}$

1x4096 1x4096 1x4096